Cory Kuiyi Gao

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Profile

Undergraduate researcher at UNC Chapel Hill working on AI safety, alignment, robustness, and AI lawmaking. My work explains the mechanisms behind vulnerabilities in advanced AI systems (agents, generative models, and memory systems) and develops evaluations, contracts enforced at run time, and alignment methods that make these systems safer and more robust. I also study how research results can be turned into lasting evidence, systems, and common practice for AI governance. Two papers at ACL 2025 (Main and Findings); two under review, one as first author (ICLR 2027) and one as co-first author.

Education

University of North Carolina at Chapel Hill Jan 2026–May 2027
Computer Science (expected major); GPA 3.68/4.00; Dean's List *Chapel Hill, NC*
Activities: Carolina AI Alignment.

The Chinese University of Hong Kong Sep 2023–May 2026
Computer Science; Faculty of Engineering ELITE Stream *Hong Kong*

Beijing No. 4 High School Sep 2020–Jul 2023

Languages. Mandarin (native), English (TOEFL iBT 110), Cantonese (intermediate).

Relevant coursework. Deep Learning (graduate); Machine Learning; Natural Language Processing; Artificial Intelligence; Algorithms and Analysis; Operating Systems; Database Systems; Formal Languages and Automata; Linear Algebra; Probability; Statistics.

Research Experience

Verified Reward Budgets for Agent Reinforcement Learning with an LLM Judge
Manuscript in preparation, first author *Chapel Hill, NC*
- **Goal:** a verification budget for agent RL computed before the run: how much verified outcome reward to buy when an LLM judge is available, with the judge trained on the same verified data so that it cannot be gamed.

Long-Term Memory for Conversational LLM Agents
StateRank, in preparation for ACL 2027, co-first author *Chapel Hill, NC*
- **Goal:** memory that keeps what a model will need from a long history and drops the rest, from reranking stored evidence (StateRank) to memory kept in the model's own weights.

Global AI Legislation Tracker
Research infrastructure, in development *Remote*
- **Built** a database of AI legislation in China, the EU, and the US from 193 primary sources, in which every statement about a statute, guideline, or court ruling cites the source line it rests on.
- **Goal:** track AI lawmaking across jurisdictions and bring technical evidence from AI systems (evaluations, run-time obligations, audit records) into legislative text and compliance practice.

RuntimeTickets: Source-Grounded Security Obligations for Agent Skills Jun 2026–present
ICLR 2027 under review, first author; with Prof. Xue (Steve) Liu and Bowei He (MBZUAI and McGill) *Abu Dhabi / Beijing*
- **Designed** a protocol in which a language model proposes task-specific security obligations for a third-party agent skill, model-free code enforces them on every outbound send, and each run leaves a record that an outside auditor can re-derive.
- **Evaluated** it on three unmodified agent frameworks against production skill scanners and runtime defenses: no injected send got through and no benign run was blocked; without it, 27.5% of attack runs succeeded.

RealGUINoise: GUI-Agent Robustness under Real-World Interface Noise Sep 2025–Sep 2026
Under review, co-first author; with Prof. Wenxuan Wang (Renmin University of China) *Hong Kong / remote*
- **Constructed** benchmark data and **built** the LLM-judge pipeline, which agrees with human labels in over 94% of cases.
- **Ran** the main experiments on 207 web, desktop, and mobile tasks with 42 interface-noise types: across 7 agent frameworks, noise lowered task success by 32.9% and safety by 14.8% (mean relative drop).

Text-KEN: Kernel-Based Novelty Scores for Text May 2025–Aug 2025
CUHK Summer Undergraduate Research Programme (SURP); with Prof. Farzan Farnia *Hong Kong*
- **Extended** the kernel entropic novelty (KEN) score from images to text and **built** the pipeline that embeds two text collections and scores how novel one is relative to the other, with no classifier to train.

Chain-of-Jailbreak: Jailbreaking Image Generation Models by Step-by-Step Editing Jun 2024–Feb 2025
Findings of ACL 2025, second author; with Prof. Wenxuan Wang and Dr. Zhaopeng Tu *Tencent AI Lab / CUHK*
- **Designed** an attack that splits a refused request into harmless-looking editing steps; it bypassed four image generation services in over 60% of cases, versus 14% for other jailbreak methods.
- **Built** CoJ-Bench and the Think-Twice Prompting defense, which blocks over 95% of these attacks.

MMSafeAware: A Benchmark for Multimodal Safety Awareness Dec 2023–Oct 2024
ACL 2025 Main, third author; with Prof. Wenxuan Wang and Dr. Zhaopeng Tu *Tencent AI Lab*

- **Constructed** image-text pairs for the benchmark (1,500 pairs, 29 safety scenarios); on it, GPT-4V judged 36.1% of unsafe inputs safe and 59.9% of benign inputs unsafe.
- **Ran** vision-centric reasoning fine-tuning, one of the three mitigations the paper tests.

Publications

[1] **Kuiyi Gao**, Xue Liu, Bowei He. (2026). *RuntimeTickets: Source-Grounded Security Obligations for Agent Skills*. **First author.** Under review at ICLR 2027. https://arxiv.org/abs/2611.03772

[2] Yongjiang Wu*, Junyuan Zhang*, Ada Chen*, **Kuiyi Gao***, Wenxuan Wang. (2026). *RealGUINoise: An Interactive Cross-Platform Benchmark for GUI Agent Robustness under Real-World Interface Noise*. **Co-first author** (*equal contribution). Under review at ACL Rolling Review. https://arxiv.org/abs/2609.38184

[3] Wenxuan Wang, Xiaoyuan Liu, **Kuiyi Gao**, Jen-tse Huang, Youliang Yuan, Pinjia He, Shuai Wang, Zhaopeng Tu. (2025). *Can't See the Forest for the Trees: Benchmarking Multimodal Safety Awareness for Multimodal LLMs*. **Third author.** ACL 2025 Main. https://aclanthology.org/2025.acl-long.832/

[4] Wenxuan Wang, **Kuiyi Gao**, Youliang Yuan, Jen-tse Huang, Qiuzhi Liu, Shuai Wang, Wenxiang Jiao, Zhaopeng Tu. (2025). *Chain-of-Jailbreak Attack for Image Generation Models via Step by Step Editing*. **Second author.** Findings of ACL 2025. https://aclanthology.org/2025.findings-acl.571/

Selected Projects

DeepCrew: Low-Cost Multi-Agent Coding under One Lead Agent Open source, 2026

- **Built** a tool in which a Claude Code or Codex lead splits a job among Claude Code agents running on DeepSeek; each agent changes only the files its task lists, and commands beyond routine work wait for approval. https://github.com/KuiyiGao/deepcrew

Selected Distinctions

Professor Joseph Lau Scholarship of Innovation, CUHK
CUHK College Entrance Scholarship
CUHK Exchange Scholarship for UNC, Spring 2026
CUHK Faculty of Engineering ELITE Stream Award

Skills

Python, PyTorch, transformers and fine-tuning, GRPO with LoRA, retrieval-augmented generation, agent tool use, runtime guardrails, automated evaluation, Docker, Playwright, Git, Linux.